Day 1: Introduction to Network ProtocolsTheory

Task 1 : Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

To capture and identify DNS and HTTPS traffic generated while accessing www.spotify.com using Wireshark.

Procedure:

1. Opened Wireshark and selected the active Wi-Fi network interface.
2. Started packet capturing.
3. Opened www.spotify.com in the web browser.
4. Waited for the website to load and allowed network communication to continue.
5. Applied the dns filter to identify DNS queries and responses.
6. Applied the tls filter to identify HTTPS-related encrypted traffic.

Observations:

1. DNS Packets:

The following DNS packets were observed:

Packet No.	DNS Query	Observation
680	HTTPS pl.scdn.co	DNS query sent
773	HTTPS pl.scdn.co	No such name response
692	A pl.scdn.co	DNS query sent
788	A pl.scdn.co	No such name response

2. HTTPS/TLS Packets:

The following TLS packets were identified:

Packet No.	Protocol	Information
352	TLS 1.3	Client Hello – www.spotify.com
355	TLS 1.3	Server Hello

405	TLS 1.3	Client Hello – open.spotify.com
408	TLS 1.3	Server Hello
587	TLS 1.3	Client Hello – apresolve.spotify.com
625	TLS 1.3	Client Hello – clienttoken.spotify.com

Analysis:

- DNS is used to resolve domain names into IP addresses or other DNS records.
- The DNS capture showed queries and corresponding responses for pl.scdn.co.
- TLS 1.3 packets showed the secure connection establishment between the browser and Spotify servers.
- Client Hello and Server Hello packets are part of the TLS handshake.
- Application Data packets indicate encrypted communication.

Result:

Successfully captured and identified DNS queries, DNS responses, and TLS 1.3 traffic related to Spotify using Wireshark.

Task 2 : Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

To perform a DNS lookup for www.flipkart.com using the nslookup command.

Command Used:

nslookup www.flipkart.com

Observed Output:

- DNS Server: dns.google
- DNS Server IP: 8.8.8.8
- Hostname: e348826.dsca.akamaiedge.net
- IP Address 1: 23.55.244.187
- IP Address 2: 23.55.244.163
- Alias: www.flipkart.com

Explanation:

The nslookup command sends a DNS query to the Google Public DNS server (8.8.8.8) to resolve the domain name www.flipkart.com.

The DNS server returns two IP addresses: 23.55.244.187 and 23.55.244.163. These addresses are associated with an Akamai CDN hostname.

The result is marked as a non-authoritative answer, which means the response was obtained from a DNS resolver's cache or another DNS source rather than directly from the domain's authoritative DNS server.

Result:
Successfully performed a DNS lookup and obtained two IP addresses for www.flipkart.com.

Conclusion:
DNS translates human-readable domain names into IP addresses, allowing browsers to locate and communicate with web servers.

Task 3 : Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

To understand the differences between HTTP and HTTPS and their role in secure web communication.

1. HTTP (Hypertext Transfer Protocol):

HTTP is a communication protocol used to transfer data between a web browser and a web server. It does not encrypt the transmitted data.

2. HTTPS (Hypertext Transfer Protocol Secure):

HTTPS is the secure version of HTTP. It uses TLS encryption to protect data exchanged between a browser and a web server.

Three Key Differences

Basis	**HTTP**	**HTTPS**
1. Security	Data is transmitted without encryption.	Data is encrypted using TLS.
2. Port Number	Uses port 80 by default.	Uses port 443 by default.
3. Data Protection	Data can be read if intercepted.	Encryption helps protect data from unauthorized reading.

Real-World Example

Example: Instagram

Instagram uses HTTPS to secure communication between users and its servers. When users log in, send messages, or browse content, HTTPS helps protect the data exchanged from unauthorized access during transmission.

Result:
Understood the three major differences between HTTP and HTTPS.

Conclusion:
HTTP is used for basic web communication, whereas HTTPS provides encrypted and more secure communication. HTTPS is essential for websites and applications that handle sensitive information.

Task 4 : Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.

To observe the sequence of DNS queries and responses when accessing a website using Wireshark.

Procedure:

1. Opened Wireshark and started capturing traffic on the active Wi-Fi interface.
2. Applied the display filter dns.
3. Opened careercenter.tops-int.com in the web browser.
4. Observed DNS packets containing Standard query and Standard query response messages.

Observations:

- Client IP Address: 10.169.37.66
- DNS Server: 8.8.8.8 (Google Public DNS)
- Website: careercenter.tops-int.com

Packet No.	**Packet Type**	**DNS Record**	**Observation**
74	Standard query	HTTPS	Query for careercenter.tops-int.com
76	Standard query	A	Query for IPv4 address
78	Standard query	AAAA	Query for IPv6 address
82	Standard query response	A	Returned IPv4 address 3.110.26.57
86	Standard query response	HTTPS	Response containing SOA information
90	Standard query response	AAAA	Response containing SOA information

Sequence of DNS Communication:

1. The browser initiated DNS queries for the domain careercenter.tops-int.com.
2. The queries were sent from the client to the DNS server at 8.8.8.8.
3. The DNS server processed the queries and sent responses back to the client.
4. The A record response provided the IPv4 address 3.110.26.57.
5. The HTTPS and AAAA queries also received responses. The displayed SOA information indicates that no matching record was returned in those responses.

Result:
Successfully captured and observed DNS queries and responses using Wireshark.

Conclusion:
DNS helps convert domain names into IP addresses. This allows the browser to locate the required server and establish communication with the website.

Task 5 : Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

Selected Protocol: DHCP (Dynamic Host Configuration Protocol)

Scenario: Connecting a Phone to Cafe Wi-Fi

Suppose I visit a cafe and connect my smartphone to its Wi-Fi network to use Instagram.

When my phone connects to the Wi-Fi, the DHCP server running on the router automatically assigns network configuration details to my phone.

These details include:

1. **IP Address:** Provides a unique address to identify my phone on the local network.
2. **Subnet Mask:** Defines the network and host portions of the IP address.
3. **Default Gateway:** Provides the address of the router through which my phone can access external networks.
4. **DNS Server:** Provides the address of the server used to resolve domain names into IP addresses.

DHCP follows a four-step process known as DORA:

- **Discover:** The phone broadcasts a request to find a DHCP server.
- **Offer:** The DHCP server offers an available IP address.
- **Request:** The phone requests the offered IP address.
- **Acknowledgement:** The DHCP server confirms the IP address assignment.

After receiving these network settings, my phone can communicate over the internet and access Instagram.

Result:
Understood how DHCP automatically assigns IP addresses and other network configuration details to devices.

Conclusion:
DHCP simplifies network connectivity by automatically providing the required network settings without manually configuring each device.